

MANUSHA DUDELA

New York, NY | [linkedin.com/in/manushadude/](https://www.linkedin.com/in/manushadude/) | manushadude@gmail.com | +1 (646) 546-1242

SUMMARY:

Results-driven BI Engineer and Data Analyst with 4+ years of experience using **SQL**, **Snowflake**, and advanced machine learning to deliver impactful data solutions across **healthcare**, **insurance**, and **fintech**. Proven ability to enhance operational efficiency, streamline workflows, and generate insights using **ETL processes**, **Data exploratory analyses**, and data visualization tools (**Tableau**, **Power BI**, **Looker**). Skilled in collaborating with stakeholders to build predictive models and dashboards, writing, tuning, and troubleshooting SQL queries, and working with Actuaries, driving improved decision-making and measurable business growth.

SKILLS:

- **Certifications:** AWS Cloud Practitioner.
- **Programming & Scripting:** Python, SQL (MySQL, PostgreSQL, Redshift), R, SAS, MATLAB.
- **Data Analysis & Modeling:** Pandas, NumPy, SciPy, Scikit-learn, OpenCV.
- **Data Visualization & BI Tools:** AWS QuickSight, Tableau, Power BI, Looker, Microsoft Excel.
- **Database Management:** SQL (Oracle, Redshift), NoSQL (MongoDB), Hadoop, Spark.
- **Cloud & DevOps:** AWS (EC2, S3, RDS), Azure, Google Cloud, Docker, Kubernetes, Terraform.
- **Analytics & Reporting:** Data Mining, Forecasting, Data Visualization, Data Modeling, KPI Analysis, Ad Hoc Analysis.

EXPERIENCE:

Data Analyst – Elevance Health Inc., New York

May 2024 - Current

- Developed and deployed data-driven solutions for **predictive analytics**, optimizing healthcare service offerings and reducing claim processing times by **17%**, leveraging **SQL** and **Python**.
- Designed interactive dashboards using **Tableau** and **Power BI** to track patient outcomes, improving decision-making for health care providers and increasing operational efficiency by **15%**.
- Automated **ETL processes** for healthcare claims data, enhancing data accuracy and reducing data retrieval time by **20%** through cloud-based platforms like **AWS** and **Snowflake**.
- Led **data exploratory analyses** for patient readmissions, decreasing hospital readmissions by **7%** using machine learning algorithms and **predictive analytics**.
- Collaborated with cross-functional teams, including product and engineering, to streamline data workflows using **Jira** within an **Agile framework**, ensuring project milestones were met on time.
- Developed healthcare compliance reports for regulatory bodies, ensuring data integrity and meeting stringent healthcare standards with 100% accuracy using **SQL** and **data visualization**.
- Collaborated with senior data analysts to refine predictive models for **insurance risk assessments**, reducing patient claims fraud by **10%** through **data mining** and **machine learning techniques**.

Associate Data Analyst – Accenture, India

Sep 2020 - Aug 2023

- Engaged with internal stakeholders to gather data requirements for patient demographics and claim analysis, ensuring alignment across departments and improving data quality by **10%** through **Salesforce CRM** and **ETL tools**.
- Analyzed and cleaned large datasets related to patient demographics, improving data quality and accuracy by **12%** using **Python** and **R** for **statistical analysis**.
- Created custom reports using **Excel** and **Power BI** to track operational metrics, reducing reporting time by **15%** and increasing visibility into healthcare service performance.
- Developed and implemented data models to assess insurance claim risks, improving claim approval efficiency by **15%** through **SQL** and **predictive analytics**.
- Designed business intelligence dashboards in **Power BI** to track customer segmentation and insurance policy performance, leading to a **12%** increase in cross-selling opportunities.
- Designed and deployed **Looker** dashboards to monitor customer segmentation and **fraud detection patterns**, reducing fraudulent claims by **10%** through **data visualization**.

EDUCATION:

Northwest Missouri State University, MO

Master of Science in Information Systems

Aug 2023 - Dec 2024

GPA - 3.60

QIS College of Engineering and Technology, India

Bachelor of Engineering in Electronics and Communication

Aug 2017 - May 2021

GPA - 3.86

ACADEMIC PROJECT: SPARSH – ML-POWERED SKIN DISEASE PREDICTION PLATFORM

- Conducted Exploratory Data Analysis on a dermatological image dataset (50,000+ images), cleaning and aggregating metadata to identify key features driving disease classification. [Skills: Python (pandas, NumPy), SQL, data cleaning, feature engineering].
- Built & Evaluated a convolutional neural network classifier—tuning hyperparameters (dropout rate, epochs) to lift accuracy to 97%—and visualized performance (confusion matrices, ROC curves) to guide model improvements. [Skills: Python (scikit-learn, TensorFlow), model evaluation, statistical analysis, data visualization (Matplotlib, Seaborn)].
- Automated ML Workflow with MLflow tracking for experiment management and parameter tuning (dropout, epochs), ensuring reproducible training pipelines and streamlined model retraining. [Skills: MLflow, TensorFlow, pipeline orchestration]